

Advanced Parasitic Capacitance Extraction using Active Learning

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Abstract—Advancements in process technology have increased the need for accurate parasitic capacitance extraction in Integrated Circuits (IC). Current rule-based extraction tools can be error-prone due to limited interconnect patterns. This study introduces a novel methodology that efficiently leverages limited training data to train the existing neural network model for parasitic capacitance extraction by applying Active Learning (AL). This methodology offers a solution to the data limitation issue while delivering results comparable to the existing model.

Index Terms—Parasitic capacitance; Parasitic extraction; Neural networks; Active learning; Gaussian Process;

I. INTRODUCTION

The evolution of semiconductor technology has escalated the number of transistors in line with Moore's law, but this growth is gradually approaching its limit. One of the key contributors to this limit is the increase of parasitic effects [1]. As process technology continues to advance, the reduction in the size of transistors and metal connections, or interconnects, has led to a significant increase in its parasitic elements, adversely impacting the performance of IC. Consequently, the accurate extraction of parasitic capacitance from the layout has become a crucial step in the IC design process. If the result of the post-layout simulation does not meet the requirements, modification is made, and simulation is repeated, a process that consumes a substantial amount of time.

The empirical method, also known as 2.5D extraction, serves as a main method for parasitic capacitance extraction. It is widely used in rule-based extraction tools, where these methods rely on matching layout patterns with appropriate, pre-characterized capacitance formulas from external libraries [2], [3], [4]. This approach offers reasonable runtime and accuracy for full-chip extraction. However, two problems arise. The first is the potential for pattern mismatches, and the second is the limited pattern coverage, both of which are pronounced in advanced nodes [2], [4]. Mismatches between layout patterns and incorrect capacitance formulas can lead to crucial inaccuracies. Additionally, limited pattern coverage can create inaccuracies, particularly in complex designs that demand multidimensional capacitance formulas.

To address the pattern mismatch and application range limitation issues due to the pattern diversity of high-density chips, a compact cross-sectional neural network model was introduced. However, due to the confidentiality of design IPs, obtaining various metal stack patterns required for training is difficult, and generating layouts, extracting metal stack patterns, and calculating parasitic capacitance also takes a significant amount of time. To address these issues, this study presents a new approach that effectively trains the neural network model with limited data by considering uncertainty estimation through AL.

II. PROPOSED METHOD

Deep Neural Networks (DNN) in parasitic extraction methodology enhance pattern coverage and reduce pattern mismatches. Such DNN models have the potential to replace hundreds of capacitance formulas, thereby providing more accurate results [5]. However, this approach requires a large number of training patterns, and the extraction or creation of these training patterns is time-consuming and resource-demanding. In this study, we propose a methodology for parasitic capacitance extraction using a DNN that is based on AL. The generation of metal pattern data is guided by design rules, and subsequently, a field solver tool is utilized to extract the parasitic capacitance values of these patterns. This strategy significantly contributes to the effective training of the DNN model, even when the number of training patterns is limited. Fig. 1 shows the overall framework.

A. Training Pattern Generation

Metal stack patterns, composed of up to five layers, are generated randomly in accordance with design rules. These patterns are designed to emulate the diversity and complexity encountered in real-world scenarios.

B. Feature Extraction

To facilitate the use of metal stack patterns as input data for both the AL model and the DNN model, composite vectors are generated. These vectors are composed of three parts, each representing the entire pattern, the aggressor metal, and the victim metal, respectively, which informs the model about the capacitance components to be extracted [5].

C. Field-Solver

Field-solvers are highly accurate and efficient techniques employed to calculate parasitic capacitances. They effectively solve the complex electrostatic problems associated with the interconnections presented in IC.

D. Active Learning

We employ a Gaussian Process Regression (GPR) model for the purpose of training pattern sampling. The extraction of

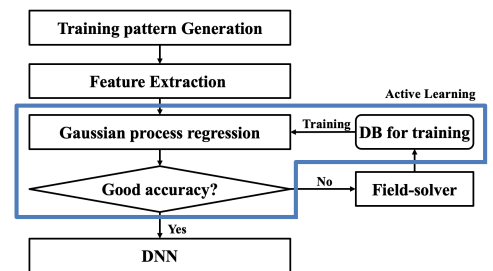


Fig. 1. Overall framework

parasitic capacitance utilizing a DNN model necessitates a substantial quantity of metal pattern data for effective training. However, the availability of actual designs for generating training data is inherently limited, and the process of extracting and creating training patterns is time-consuming. To address this, we utilize the trained Gaussian Process (GP) to measure the uncertainty of the data. Subsequently, we apply AL techniques to sample data characterized by high uncertainty. This approach allows us to significantly reduce the volume of metal pattern data required for training the DNN model. The posterior mean and variance of the GP are derived by applying conditional probabilities as follows:

$$\mu_* = k(X_*, X)^\top (k(X, X) + \sigma^2 I)^{-1} Y,$$

$$\sigma_*^2 = k(X_*, X_*) - k(X_*, X)^\top (k(X, X) + \sigma^2 I)^{-1} k(X, X_*), \quad (1)$$

where the new input values for a prediction are represented by X_* , while the previous inputs are denoted by X . The predicted value of the objective function for the new input is symbolized by μ_* , and σ_*^2 signifies the uncertainty associated with the prediction. The previously observed output values are represented by Y , and $k(X_*, X)$ stands for the covariance matrix. The GP formula not only provides the predicted value but also quantifies the level of confidence associated with the prediction. The regions with the largest confidence intervals correspond to the most uncertain areas of the prediction model. By strategically sampling and utilizing data, we can effectively minimize the quantity of training data required. The parameters of the GP used for AL are optimized using negative log-likelihood and the Adam optimizer [6].

E. DNN Model

A DNN model is implemented with the aim of predicting the parasitic capacitance of 2D cross-sectional patterns [5]. This model is composed of a total of five layers, utilizing ReLU and hyperbolic tangent as activation functions. The model is trained using the Adam optimizer [6]. Kernel initialization is performed using HeNormal and GlorotNormal. The cost function utilizes Mean Square Error (MSE), and to achieve a higher degree of accuracy, the input data is normalized to a value with zero mean and a standard deviation of 1 [7].

III. EXPERIMENTAL RESULTS

In this study, the generation of training patterns was executed, predicated on the 32/28 nm process, employing a total of 30K metal stack patterns. *Synopsys Raphael 2021.06 rc2* field-solver was used to extract the parasitic capacitance values of both the training and test patterns. GPR model was implemented using the *PyTorch 2.0.1* library, while the DNN model was developed with the *TensorFlow 2.12.0* library. For comparative analysis, an experiment was conducted using the conventional DNN model alone, wherein 80% (24K) of the total 30K patterns and 12K were allocated for training. The GPR model was employed to sample a total of 12K and 7.5K training patterns, which were subsequently used as the training patterns for the DNN model. The DNN models trained with 24K and 12K patterns exhibited MSEs of 0.054 and 0.09, respectively, with the proportion of outliers exceeding 5% being 1.7% and 2.6%. Conversely, the DNN models trained with 12K and 7.5K patterns sampled via GPR showed MSEs of 0.059 and 0.087, respectively. The proportion of outliers exceeding 5% were 2.4% and 3%, respectively. All models were tested with the same set of

TABLE I. COMARISON OF RESULT BETWEEN ACTIVE LEARNING WITH DEEP NEURAL NETWORKS AND CONVENTIONAL DEEP NEURAL NETWORKS

	Method			
	<i>DNN [5]</i>	<i>Ours</i>	<i>DNN [5]</i>	<i>Ours</i>
# of training datasets	24K(0%)	12K(50%)	12K(50%)	7.5K(68%)
Mean square error	0.054(0%)	0.059(9%)	0.09(66%)	0.087(61%)
Outliers > 5% error	1.7%(0% p)	2.4%(0.7% p)	2.6%(0.9% p)	3%(1.3% p)

patterns, which constituted 20% (6K) of the total 30K patterns. The results were calculated by taking the average after 10 experiments. Table I shows the accuracy of the four methods. In the table, a DNN trained on a 24K training dataset is used as the baseline. Red indicates an increase rate, while blue indicates a decrease rate.

IV. CONCLUSION

This study introduces a novel modeling methodology that leverages AL for the sampling of training patterns, subsequently training a DNN to extract parasitic capacitance. This methodology resolves the problem of significant computational overhead associated with generating training metal patterns in conventional DNN-based parasitic capacitance extraction methods. By employing this approach, it accomplishes precise forecasting of parasitic capacitance values while utilizing fewer number of metal patterns. In this study, we observed a slight performance difference between the conventional DNN model trained with 24K datasets and the DNN model trained with 12K datasets sampled using GPR. Furthermore, when compared with a DNN model trained with an equivalent number (12K) of randomly selected datasets, we noted a 34% reduction in the MSE and a 7% decrease in the ratio of outliers exceeding 5%. Additionally, our method demonstrated comparable performance to a DNN model trained with 12K randomly selected datasets, but only using 7.5K datasets. These results serve to underscore the efficacy and efficiency of the proposed methodology.

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